

The Combined Effects of Dietary Fat and Estrogen on Survival, 7,12-Dimethylbenz(a)anthracene-Induced Breast Cancer and Prolactin Metabolism in Rats^{1,2}

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ABSTRACT The relationships between dietary fat concentration (10 or 40% of energy), fat source (corn oil or beef tallow) and estrogen (control, ovariectomy or ovariectomy with estrogen replacement) to 7,12-dimethylbenz(a)anthracene (DMBA)-induced breast carcinogenesis and survival in rats were studied in a 2 × 2 × 3 factorial experiment. Female Sprague-Dawley rats given DMBA (2.5 mg/100 g body wt, intragastric) at 55 d of age were randomly allocated to three groups 48 h later: sham ovariectomy (control), ovariectomy (OVX) or ovariectomy with a subcutaneous estrogen implant (OVX+E). Each group was subdivided into dietary groups fed 10 and 40% of energy as corn oil or beef tallow for 70 wk. OVX+E rats exhibited serum estrogen concentrations in excess of physiologic values. Survival at 70 wk for the 3 hormonal groups was control 51%, OVX 67% and OVX+E 13%. Mortality in controls was doubled by feeding a high fat diet; no diet effect was detected in OVX or OVX+E rats. Palpable tumors developed in 74, 14 and 60% of control, OVX and OVX+E rats, respectively. High fat diets approximately doubled the hazard of developing a palpable tumor. Adenocarcinoma prevalence was 58, 12 and 63% in control, OVX and OVX+E rats, respectively. The odds of having any tumor, an adenocarcinoma or an adenoma were multiplied by 3.6, 2.8 and 2.3, respectively, for rats fed high vs. low fat. Additional studies showed that diet had no effect on serum prolactin or estrogen concentrations or metabolism and clearance of intravenously administered radiolabeled prolactin. We demonstrated that high dietary fat concentration enhances breast carcinogenesis independently of cyclic ovarian function, although the presence of estrogen may be a prerequisite for significant dietary modulation. The effect of fat on breast cancer is not mediated by major changes in systemic prolactin metabolism. *J. Nutr.* 125: 1192-1204, 1995.

INDEXING KEY WORDS:

- rats • estrogen • prolactin
- dietary fat • breast cancer

Raising the concentration of dietary fat increases the incidence and multiplicity of mammary cancers in a wide variety of carcinogen-induced, radiation-induced, spontaneous, transplantable and metastatic rodent models (Carroll and Kohr 1971, Freedman et al. 1993, Tannenbaum 1942, Welsch 1992). The relevance of these observations to human cancer has been strengthened by recent reports describing increased growth rates of human breast carcinoma transplants in immunodeficient mice fed high fat diets (Blank and Ceriani 1989, Borgeson et al. 1989, Gabor et al. 1990, Gonzalez et al. 1991). Despite these extensive studies, the precise mechanisms whereby fat enhances experimental mammary carcinogenesis remain obscure (Welsch 1987 and 1992).

Ovarian hormones play a central role in the development and differentiation of mammary glands, and in the etiology and progression of mammary tumors (Welsch 1985). The possibility that estrogen mediates the effects of dietary fat on rodent mammary cancer has been a popular hypothesis investigated by several approaches (Carroll and Kohr 1975, Welsch 1985, 1987 and 1992). Rats treated with an anti-estrogen drug continued to show a higher incidence of tumors when fed a high fat diet although

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overall tumor risk was substantially reduced (Chan et al. 1975). However, these provocative results must be interpreted with caution because most anti-estrogen pharmaceuticals have mixed agonist and antagonist effects on estrogen responsive tissues and few of the anti-estrogens have been evaluated in studies of rats fed diets varying in fat (Chan et al. 1975, Helgason et al. 1982). Rats ovariectomized during the late promotion phase of 7,12-dimethylbenz(a)anthracene (DMBA)⁴ carcinogenesis subsequently showed a greater incidence of estrogen independent mammary tumors when fed a high fat diet (Cohen et al. 1981). Studies of serum estrogen concentrations in rats fed diets varying in fat level and source have produced variable results (Chan et al. 1977, Ip and Ip 1981, Welsch 1987, Wetsel et al. 1984). However, the majority of carefully controlled studies have failed to demonstrate modulation of serum estrogen concentration by dietary fat between 10 and 40% energy intake (Ip and Ip 1981, Welsch 1987, Wetsel et al. 1984). One of our goals was to investigate the role of functioning ovaries in the modulation of mammary carcinogenesis by dietary fat. We therefore examined these relationships in rats whose cyclic ovarian function was eliminated by ovariectomy. The ovariectomized rats were divided into two groups and their estrogen levels were maintained throughout their lifespans by controlled-release estrogen implants or empty implants. An important feature of the present study was the evaluation of tumorigenesis over 70 wk rather than the 20–35 wk interval usually used. This allowed us to determine the effects of dietary fat concentration and source on survival in addition to tumor incidence.

Prolactin also regulates the growth and differentiation of normal and malignant mammary tissue (Welsch and Nagasawa 1977). Hyperprolactinemia produced by administration of prolactin (Meites et al. 1971), transplantation of anterior pituitaries (Welsch et al. 1970), electrolytic lesions of the median eminence of the hypothalamus (Welsch et al. 1970) or by pharmacologic agents (Manni et al. 1977) enhances the incidence or growth rate of mammary cancers in experimental models. Several studies have examined the relationship of dietary fat to serum prolactin concentrations and pituitary prolactin secretion (Chan et al. 1975 and 1977, Clinton et al. 1985, Hopkins et al. 1981, Ip et al. 1980, Wetsel et al. 1984). The earlier evidence from rodent studies suggesting that diets high in fat were associated with greater concentrations of immunoreactive serum prolactin (Chan et al. 1975 and 1977) was not confirmed in subsequent experiments (Clinton et al. 1985, Hopkins et al. 1981, Wetsel et al. 1984). However, recent studies indicate that prolactin is synthesized and secreted in multiple forms (Sinha 1992). Furthermore, the systemic metabolism of prolactin to different forms produces significant changes in the biological activity and in the

detection of prolactin by immunoassays due to alterations in antibody binding (Sinha 1992). Studies by Sinha and Baxter (1979) have suggested that mouse strains exhibiting greater mammary tumor risk exhibit alterations in prolactin metabolism. The possibility that dietary fat may modulate the systemic metabolism of prolactin, thereby altering its bioactivity or immunoreactivity, has not been fully explored. The studies we report address the effects of dietary fat concentration and source on the systemic metabolism of exogenously administered prolactin.

MATERIALS AND METHODS

Animals. Female Sprague-Dawley rats (Harlan Industries, Madison, WI) were individually housed in stainless steel wire-bottomed cages in rooms with air temperature maintained between $24 \pm 2^\circ\text{C}$, with 14 h of light per 24 h cycle. The care and use of laboratory animals followed the guidelines set forth by the University of Illinois and the *Public Health Service Policy on Humane Care and Use of Laboratory Animals*, revised September 1986.

Diets. The AIN-76A diet (AIN 1977 and 1980) was fed to all rats before their assignment to experimental diets (Table 1) that varied in lipid source (beef tallow or corn oil) and the percentage of energy derived from fat (10 and 40%). The ratios of protein, vitamins, minerals, choline and cellulose to energy were constant. Diets prepared at monthly intervals (Teklad, Madison, WI) were analyzed for fat concentration upon arrival (Folch et al. 1957), and stored at -20°C until fed. Samples of the beef tallow and corn oil were analyzed (Johnston 1971) for fatty acid content (Table 1). Distilled water and fresh food were provided three times weekly. Total food intake over the entire study was calculated for each rat by summation of the measured food intake from each feeding. We calculated total energy intake for each rat using conversion factors of 15.5 and 18.7 kJ/g for the high and low fat diets, respectively (Table 1). Total energy intake for each rat was divided by the number of days the rat survived in order to establish the mean daily energy intake. These numbers were used for descriptive analysis.

Survival and mammary cancer study. Three hundred eighty-four female Sprague-Dawley rats, averaging 48 d of age, were fed the AIN-76A diet. After 1 wk of adaptation, food jars were removed at 0600 h and 2.5 mg DMBA/100 g body wt (Kodak, Rochester, NY) was administered to each rat between

⁴Abbreviations used: control, sham ovariectomy; DMBA, 7,12-dimethylbenz(a)anthracene; OVX, ovariectomy; OVX+E, ovariectomy with estrogen implant.

TABLE 1
Composition and nutrient density of the diets

	Corn oil		Beef tallow	
	Low	High	Low	High
Percentage composition, g/100 g				
Casein ¹	17.24	20.85	17.24	20.85
DL-Methionine	.30	.36	.30	.36
Corn oil ²	4.00	20.00	1.00	1.00
Beef tallow ³	0.00	0.00	3.00	19.00
Carbohydrate ⁴	68.95	47.53	68.96	47.53
Cellulose	5.00	6.03	5.00	6.03
Mineral mix ⁵	3.50	4.23	3.50	4.23
Vitamin mix ⁵	1.00	1.20	1.00	1.20
Nutrient density				
kJ/g ⁶	15.5	18.7	15.5	18.7
Percentage of energy as lipid	10	40	10	40

¹Vitamin-free casein, 87% protein.

²Beef tallow fatty acid composition: 14:0 4%, 14:1 1%, 16:0 34%, 16:1 4%, 18:0 20%, 18:1 32%, 18:2 2%, 18:3 1%, all others <1%.

³Corn oil fatty acid composition: 16:0 14%, 18:0 5%, 18:1 21%, 18:2 56%, 18:3 2%, all others <1%.

⁴Carbohydrate fraction composed of dextrin (66%), sucrose (16.5%) and expanded maltose-dextrin (16.5%).

⁵Vitamin and mineral mixtures according to the American Institute of Nutrition formulation for rats and mice (AIN 1977 and 1988).

⁶Calculations based upon estimates of metabolizable energy of 16.74 kJ/g of protein and carbohydrate and 37.66 kJ/g of lipid.

1000 and 1200 h in 0.3 to 0.4 mL of corn oil by intragastric intubation. Food jars were returned to the cages at 1600 h. Forty-eight hours after DMBA administration, the rats were randomly assigned to one of three groups: ovariectomy, with a silastic implant (5 mm in length, 1.57 mm i.d., 3.18 mm o.d., Dow Corning, Midland, MI) containing beta-estradiol-3-benzoate (Sigma Chemical, St. Louis, MO) placed subcutaneously on the back (OVX+E); ovariectomy with an empty implant (OVX); or sham ovariectomy with an empty implant (control). The implants (Legan et al. 1975) were designed to maintain a serum estrogen concentration within the physiologic range (Kao and Ramirez 1979). Forty-eight hours after surgery, rats within each hormonal treatment were randomly subdivided among the four dietary treatments with 32 rats assigned to each of 12 treatment groups. Ketamine (Bristol Laboratories, Syracuse, NY) was the anesthesia for all surgical procedures. All implants were replaced after 35 wk.

Each rat was examined weekly for palpable tumors beginning 5 wk after DMBA administration. Moribund rats were necropsied, and their tumors were placed in 10% buffered formalin for subsequent histopathologic studies. All survivors were necropsied 70 wk after DMBA administration.

The effectiveness of randomization to the 12 treatment groups was evaluated by one-way ANOVA of initial body weights. To examine the effects of treatments on growth before the appearance of cancer-associated morbidity, body weights at 28 wk were analyzed by a three-way factorial ANOVA.

Ninety-three percent of rats remained alive at 28 wk, with the majority in good health. Analytic models included fat level, source, ovarian status and their interactions.

Time to first palpable tumor, and time to death, were studied separately using several techniques of survival analysis. Kaplan-Meier (Lee 1980) survival plots for each of the 12 hormone by diet treatment groups did not suggest a single parametric model compatible with data for all twelve treatment groups. Nonparametric proportional hazards modeling (Cox 1972, Harrell 1986b) was then used in circumstances where the underlying assumption that hazard ratios of different groups are constant over time seemed tenable. Where hazard ratios varied according to some design factors but not others, partial (nested) proportional hazards models were used, with tests of parameters across factor levels undertaken using Wald statistics (Anonymous 1988). When even partial proportional hazards modeling was inappropriate, the log-rank test was employed to document average systematic effects of one variable, controlling for levels of other variables by appropriate stratification.

The relationships of diet and ovarian status to tumor prevalence at necropsy were analyzed using factorial multiple logistic regression analyses (Harrell 1986a, Hosmer and Lemeshow 1989) of occurrence of any tumor, adenocarcinoma or adenoma. Likelihood-ratio tests were used for model fitting, and Wald statistics for testing specific experimental factors using parameters of fitted models. Mean tumor counts in tumor-bearing rats were studied using

weighted least-squares factorial modeling (any tumors, adenocarcinomas) or, when numbers of animals with tumors were insufficient to permit this (adenomas), by extended Mantel-Haenszel testing of average effects of each factor (estrogen supplementation, fat level, fat source), adjusted for the remaining two (Koch et al. 1985).

Basal hormone study. Seventy-two rats were randomly assigned to the 12 treatment groups according to the protocol used in the mammary cancer study, except that DMBA was not administered. Twenty-four additional rats were randomly allocated to the four diets in the control group. We examined serum and pituitary hormone concentrations in rats at 24 wk of age, 16 wk after initiation of dietary treatments. When the control rats were clearly in diestrus or proestrus, as determined by vaginal smears, they were decapitated between 1500 and 1600 h. Trunk blood was collected and allowed to clot for 4 h at room temperature, and serum was separated by centrifugation. The anterior pituitary glands were removed, weighed, homogenized and centrifuged in 0.01 mol/L phosphate-buffered saline. Serum and pituitary supernatants were stored at -20°C until assayed. Prolactin was measured by the double-antibody method with materials supplied by the National Institutes of Health (Bethesda, MD). ^{125}I (New England Nuclear, Boston, MA) was used for radio-iodination of rat prolactin. The primary antibody (rabbit) was NIAMDD-anti-rPRL-S-8. Goat anti-rabbit IgG (Research Products, Elk Grove Village, IL) was used as the secondary antibody. The amount of prolactin was expressed relative to NIAMDD-rat-PRL-RP-2. Standard curves were completed in triplicate for each assay. Total estrogen concentrations were determined with a double antibody RIA kit (Radioassay Systems Laboratories, Carson, CA). Effects of fat level, fat source and estrogen status on serum and pituitary hormone concentrations were evaluated by three-way ANOVA, and Scheffe's one-way multiple-comparison method was subsequently used to test differences among hormone groups pooled across diets (Steel and Torrie 1980).

Prolactin metabolism study. Seventy-two rats were randomly assigned to the 12 treatment groups according to the protocol followed in the mammary cancer study, except that DMBA was not administered. Growth and feed intake were recorded. After 16 wk, at 0800 h each rat received an intraperitoneal injection of 2-Br- α -ergocryptine (CB-154, 0.5 mg/kg body weight, Sandoz, Ltd., East Hanover, NJ) dissolved in ethanol-saline, to suppress endogenous prolactin release. At 1000 h a jugular catheter was inserted in each rat after administration of Ketamine anesthesia. At 1200 h a 0.2-mL mixture of 0.5 μg immunoreactive rat prolactin and 25 ng of ^{125}I -prolactin (NIAMDD-rPRL-I-5) was administered. The prolactin had been iodinated with ^{125}I by the

Chloramine-T method (Greenwood et al. 1963). At 10 and 20 min following injection, 1.0-mL samples of blood were withdrawn via the cannula and serum was obtained. The 500- μL serum samples were fractionated with Sephadex G-100 columns (Sinha and Baxter 1979) and 70 1-mL fractions were collected and analyzed for radioactivity and RIA as previously described (Clinton et al. 1985) using NIAMDD-RP-2 as the standard. Because ^{125}I was used for labeling of prolactin before injection, the radioactive antigen for prolactin assays was prepared with ^{131}I to allow discrimination against the fraction of prolactin labeled with ^{125}I . The areas under the peaks for absolute radioactivity and immunoreactivity were calculated and expressed as the percentage of total activity. The free iodine peak was not included in calculating the total activity.

The proportions of prolactin in each peak at 10 and 20 min were analyzed separately for radioactivity and by RIA, using repeated measures ANOVA that included fat concentration and source, estrogen status, sampling time and their interactions in the statistical models (Bock 1975, SAS 1985). Total prolactin was similarly analyzed for time, dietary and hormonal effects.

RESULTS

Body weight and food intake. At randomization, differences in mean body weights among the 12 groups were compatible with chance ($P = 0.45$). Hormonal treatment significantly affected body weight at 28 wk (Fig. 1), with control weights intermediate between the elevated and depressed weights of the OVX and OVX+E rats, respectively ($P < 0.0001$ for main effect). This response was seen in every dietary group. An effect of fat concentration was also noted, with high fat-fed rats being heavier than those fed low fat diets. The effect of fat level on body weight in OVX+E rats was approximately half that in the control and the OVX groups ($P < 0.0001$ for fat level main effect, $P = 0.002$ for interaction). No effect of fat source was found among high fat-fed animals, but rats fed low fat corn oil diets were heavier at 28 wk than those fed low fat beef tallow diets ($P = 0.007$ for fat source main effect, $P = 0.002$ for interaction).

Mean daily energy intakes averaged over the entire experiment, shown in Figure 2, generally reflected differences found in 28-wk body weights. Because survival times varied markedly across treatment groups, and these values treat the mean daily intakes of rats who died early and those of rats surviving 70 wk identically, they should receive limited emphasis. To avoid overinterpretation, we have refrained from formal statistical analysis and inference.

Survival. Figure 3 shows survival plots for dietary fat concentration within the fat source and hormonal

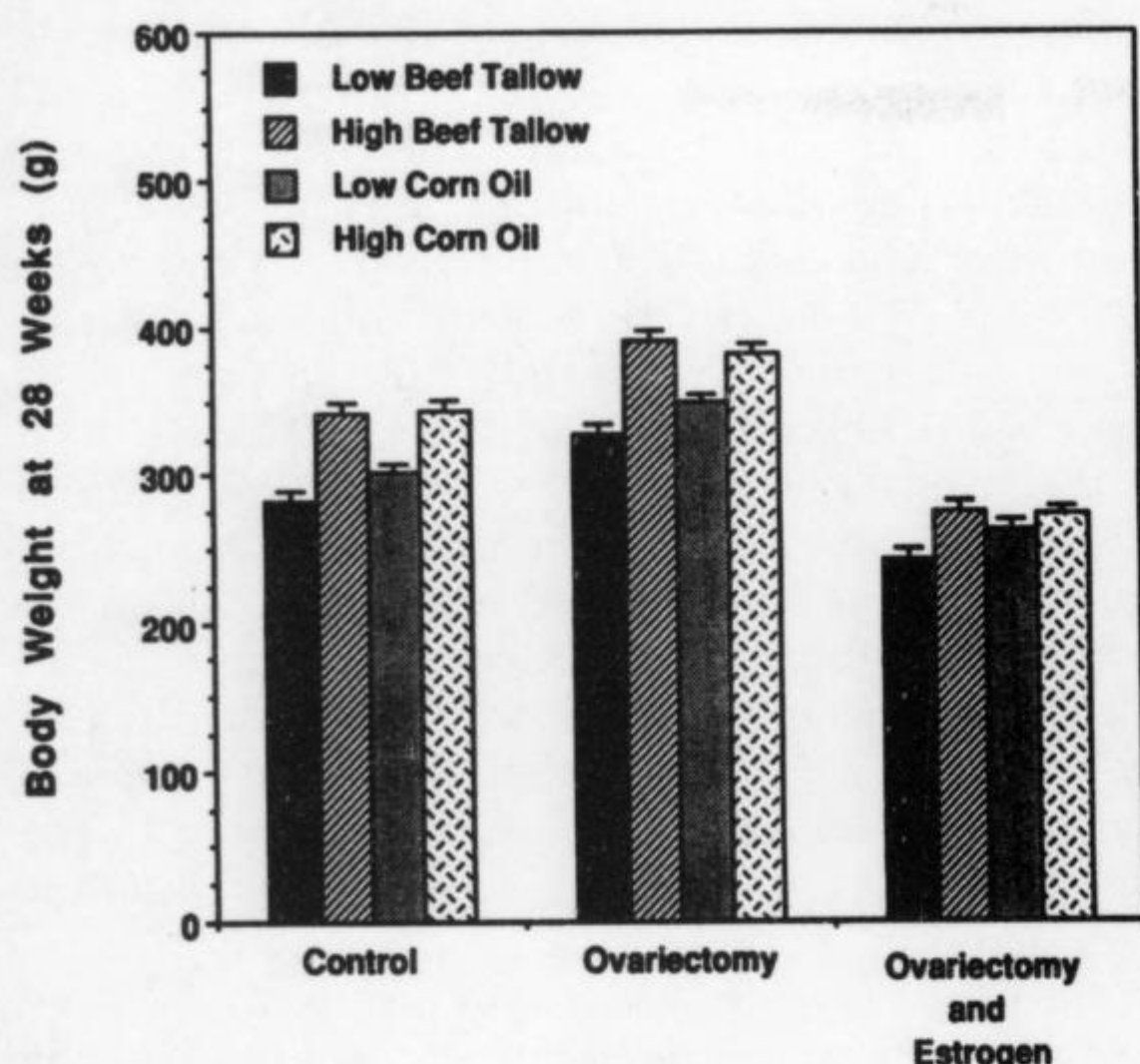


FIGURE 1 Body weights for female Sprague-Dawley rats from three experimental groups [sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E)] fed diets varying in dietary fat concentration and source at 28 wk following DMBA administration. Mean initial body weights for treatment groups ranged from 168 to 174 g. Values represent means $\pm$ pooled SEM. Weights of controls were intermediate between the elevated and depressed values of the OVX and OVX+E rats, respectively ($P < 0.0001$, for main effect). High fat-fed rats weighed more than those fed low fat diets ($P < 0.0001$, for main effect). The effect of fat concentration on body weight in OVX+E rats was approximately half that in the control and the OVX groups ($P < 0.002$ for interaction). No effect of fat source was found among the high fat-fed rats, but those fed low fat corn oil diets weighed more than those fed low fat beef tallow diets ($P = 0.007$ for fat source main effect, $P = 0.002$ for interaction).

subgroups. Observed 70-wk survival proportions for the hormonal groups, pooled across diets, were 67% for OVX, 51% for control and only 13% for OVX+E. This reflects severely compromised survival of the OVX+E rats relative to either the controls or the OVX rats ($P < 0.0001$ for each pairwise stratified log-rank test). Ovariectomized rats showed a modest enhancement of survival compared with controls (stratified log rank $P = 0.01$).

The effects of dietary fat concentration on survival depended upon the hormonal treatment (Table 2). Among controls, fat source had no effect, and fat level had no apparent effect on survival through 55 wk. However, after 55 wk the situation changed, and by termination of the experiment at 70 wk, high fat was associated with doubled mortality (66 vs. 33%, $P = 0.003$, pooled over fat source). On the other hand, no significant effect of fat level was found by joint proportional hazards modeling of the OVX and OVX+E groups. Fat source was influential only in

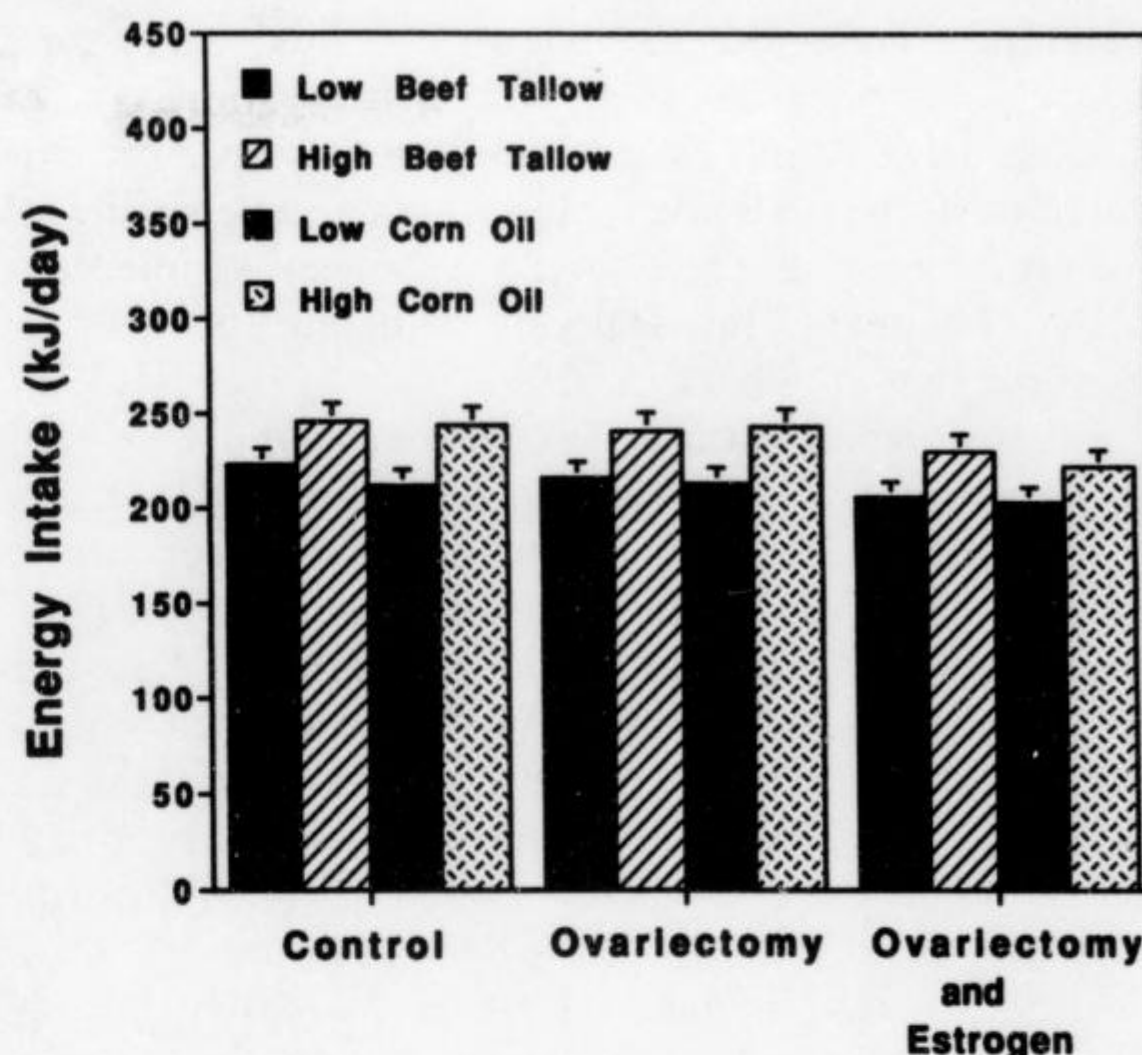


FIGURE 2 Daily energy intake for female Sprague-Dawley rats from three experimental groups [sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E)] fed diets varying in dietary fat concentration and source for 70 wk following DMBA administration. Values represent means $\pm$ pooled SEM. High fat-fed rats averaged 25 kJ/day higher consumption than low fat-fed rats, and OVX+E rats averaged 15 kJ/d lower consumption than the other groups.

OVX rats, with those fed corn oil experiencing earlier mortality than those fed beef tallow (hazard ratio $\pm$ SE, 2.7 ± 0.9). No effect of fat source was found among OVX+E rats (interaction $P = 0.11$).

TABLE 2

The effects of dietary fat and hormone treatment on survival of 7,12-dimethylbenz(a)anthracene-treated rats after 70 wk¹

Diet	Control	OVX	OVX+E	Pooled
% of rats surviving at 70 wk ²				
Beef tallow				
Low	72	78	16	55
High	31	81	9	41
Corn oil				
Low	63	63	19	48
High	38	47	10	32
Pooled	51	67	13	44

¹Dietary fat, as either beef tallow or corn oil, was provided at 10 or 40% of energy. Hormone treatments were sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E). $n = 32$ rats per group.

² $P < 0.0001$ for OVX+E vs. OVX or vs. controls, $P = 0.01$ for OVX vs. controls, $P = 0.002$ for high vs. low fat among controls and $P = 0.003$ for beef tallow vs. corn oil among OVX rats, based on comparisons of entire survival curves.

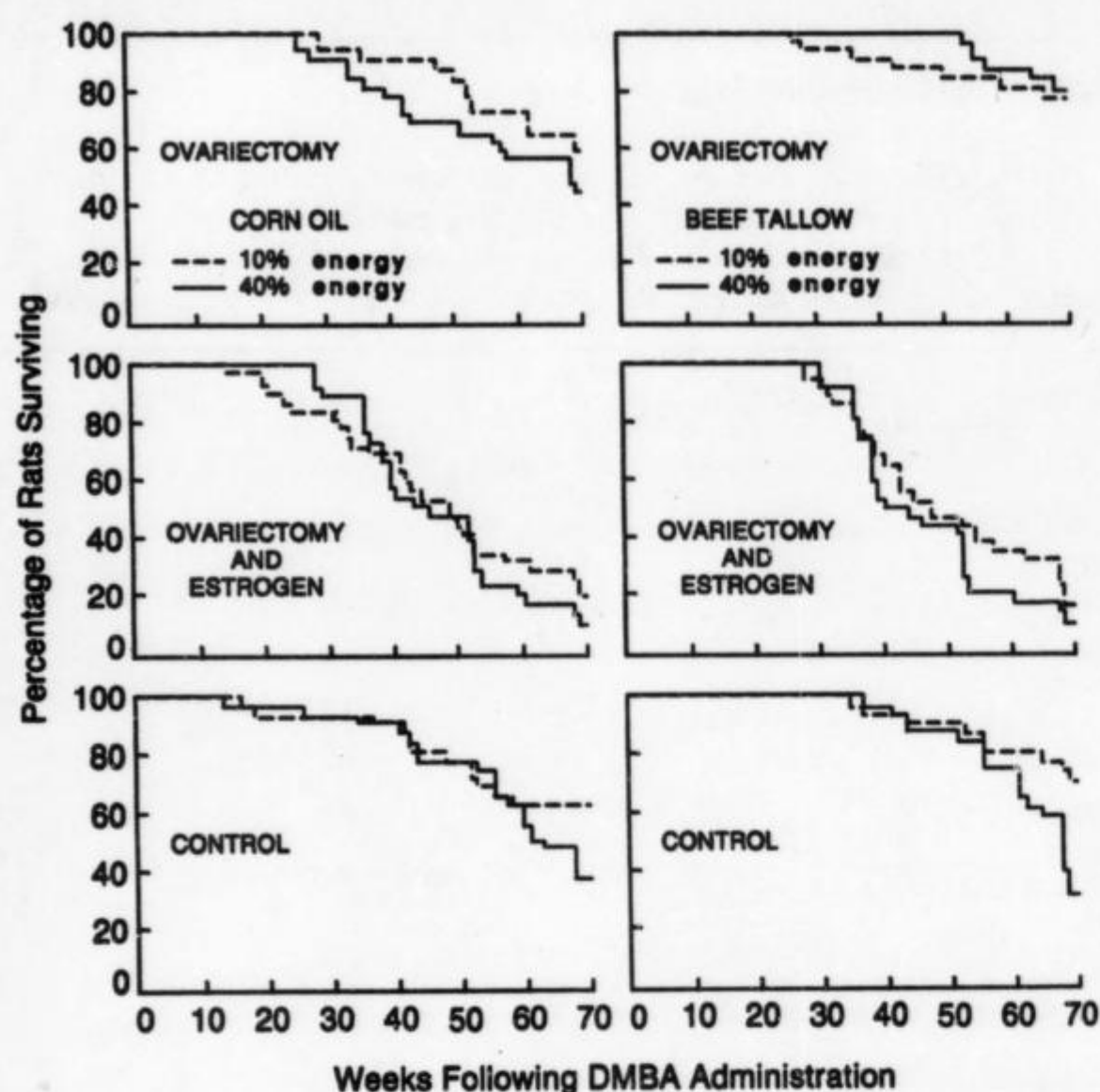


FIGURE 3 The effects of dietary fat concentration and fat source on survival of female Sprague-Dawley rats from three experimental groups [sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E)] treated with DMBA at wk 0. Observed 70-wk survival proportions for the hormonal groups, pooled across diets, were 67% for OVX, 51% for controls and 13% for OVX+E. Ovariectomized rats survived longer than controls ($P = 0.01$), and both groups survived much longer ($P < 0.0001$) than the OVX+E rats. High dietary fat decreased survival significantly only in controls. By the termination of the study, controls fed high fat diets exhibited a doubled mortality rate (66 vs. 33%, $P = 0.003$, pooled over fat source). Dietary corn oil decreased survival compared with dietary beef tallow significantly only in OVX rats ($P = 0.003$).

Palpable tumors. Five hundred sixty-four breast tumors were detected by palpation in the 190 rats that developed tumors. Table 3 gives percentages of rats developing palpable tumors, and total numbers of tumors palpated by 70 wk. Figure 4 shows Kaplan-Meier plots of time to first tumor palpation, upon which statistical tests were based. Ovariectomized rats were highly resistant to tumorigenesis compared with the other two groups ($P < 0.0001$, for log-rank tests of OVX vs. controls, vs. OVX+E and vs. pooled data from both groups). The effect was present regardless of dietary fat level or fat source. No significant difference was found between the controls and the OVX+E rats. The effect of fat level was consistent with a stratified proportional hazards model, with strata formed by joint hormone and dietary source combinations, and constant hazard ratio $\pm$ SE of 2.1 ± 0.3 for high vs. low fat-fed rats ($P < 0.0001$ by likelihood-ratio test). Fat source did not consistently delay or retard development of palpable tumors, but displayed a more complex relationship to tumor ap-

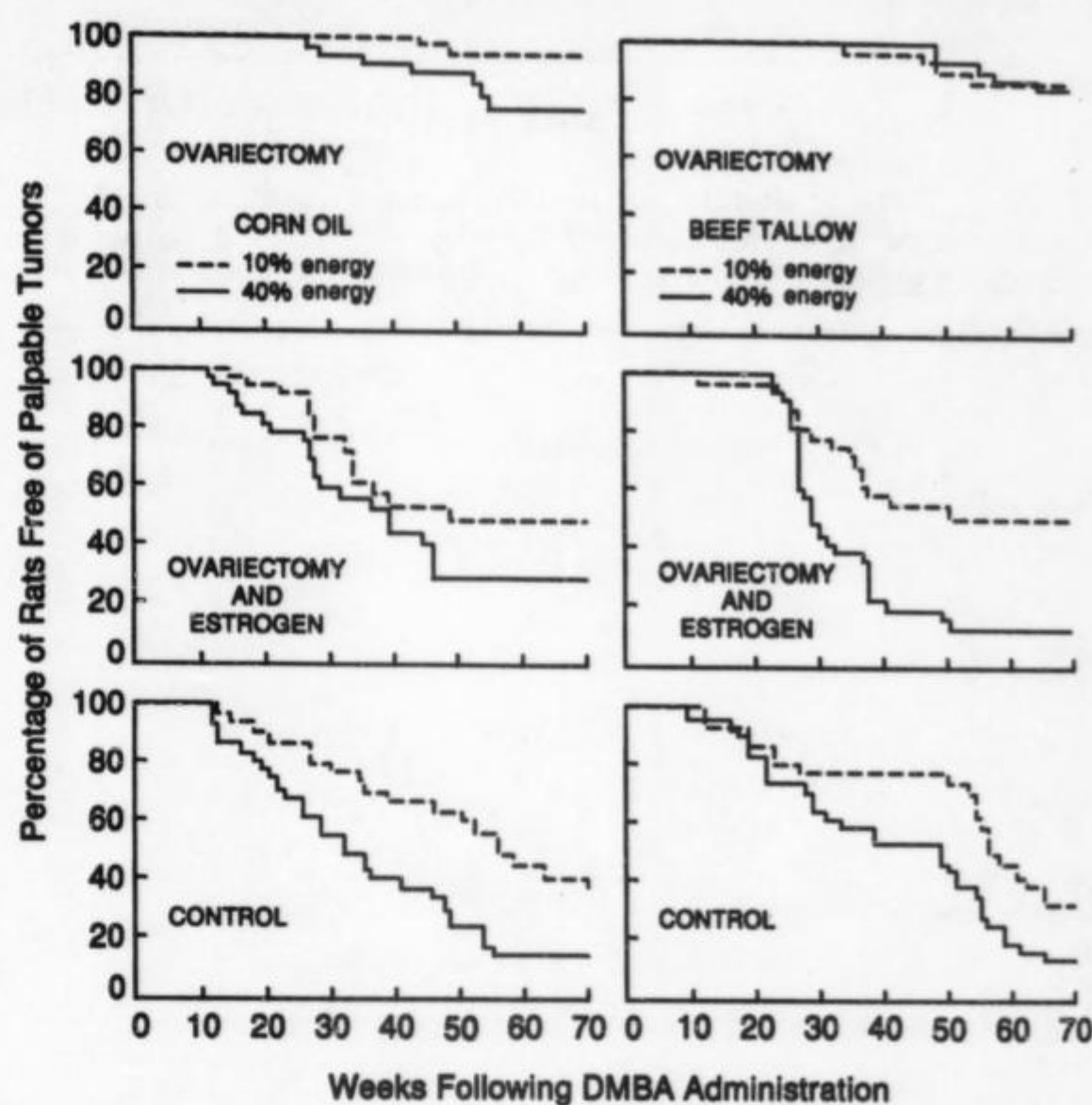


FIGURE 4 The effects of dietary fat concentration and fat source on palpable mammary tumor incidence in female Sprague-Dawley rats from three experimental groups [sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E)] treated with DMBA at wk 0. Ovariectomized rats exhibited fewer palpable tumors than controls or OVX+E ($P < 0.0001$). No significant difference in palpable tumor incidence was noted between controls and OVX+E. The ratio of palpable tumor hazards for high vs. low fat-fed rats was 2.1 ± 0.3 ($P < 0.0001$). Fat source generally had little effect.

pearance in OVX+E rats fed high fat diets. Among the rats fed beef tallow diets who developed palpable tumors, 80% first displayed a tumor in a 12-wk period, between 26 and 37 wk after DMBA administration, with roughly 10% of tumors found both earlier and later. In contrast, only 38% of comparable rats fed corn oil diets first displayed a palpable tumor in wk 26–37. The first palpable tumor in the middle 80% of such rats displaying tumors was spread evenly over wk 15–46. The extremely narrow concentration of palpation times in rats fed beef tallow vs. those fed corn oil accounted for failure of stratified proportional hazards modeling for fat source (lack of fit $P = 0.002$). Note that until wk 25 of this study, OVX+E rats fed high fat diets from corn oil showed more palpable tumors than comparable rats fed beef tallow, but by wk 35 this relationship was reversed. Such contrasting patterns of tumor development may make results of studies of fat source in hormonally altered rats strongly dependent on length of the promotion interval. Consistent with our previous observations (Clinton et al. 1984, 1986 and 1988), a modest percentage (7%) of all palpable tumors regressed before necropsy. No significant effects of treatment were observed on tumor regression.

TABLE 3

The effects of dietary fat and hormone treatment on the percentage of female Sprague-Dawley rats with a palpable mammary tumor and the total number of palpable tumors¹

Diet	Control	OVX	OVX+E	Pooled
% of rats with a palpable tumor ^{2,3}				
Beef tallow				
Low	69	13	47	43
High	88	16	88	64
Corn oil				
Low	56	6	44	35
High	84	22	62	56
Pooled	74	14	60	50
total number of palpable tumors ^{2,3}				
Beef tallow				
Low	48	4	42	94
High	105	8	88	201
Corn oil				
Low	43	4	48	95
High	91	9	74	174
Pooled	287	25	252	564

¹Dietary fat, as either beef tallow or corn oil, was provided at 10 or 40% of energy. Hormone treatments were sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E). *n* = 32 rats per group.

²Including those showing a palpable tumor that subsequently regressed.

³*P* < 0.0001 from log-rank tests of OVX vs. controls or vs. OVX+E, and for estimated tumor hazard ratio of 2.1 ± 0.3 for high vs. low fat-fed rats. Fat source generally had little effect.

Tumor prevalence at necropsy. Seven hundred mammary tumors were found at necropsy, with 67% showing histologic features consistent with adenocarcinoma (Table 4) and the remainder classified as adenoma and fibroadenoma (32%), or fibrosarcoma (1%). All measures of tumor occurrence were dramatically lower in the OVX rats and differences among dietary treatments could not be adequately evaluated in this group. Thus, formal statistical evaluation was confined to the control and OVX+E rats. Multiple logistic modeling showed no indication of an effect by fat source on final tumor prevalence. However, fat level had significant effects on the prevalence of adenocarcinomas (*P* < 0.0001) and of adenomas (*P* = 0.01). The odds of developing any tumor, an adenocarcinoma or an adenoma were increased by factors of 3.6 ± 1.0, 2.8 ± 0.8 or 2.3 ± 0.7 (mean ± SE), respectively, in animals fed the high fat diets. Among rats showing at least one adenocarcinoma at necropsy, we observed no dietary effects on adenocarcinoma multiplicity. With respect to adenomas (Table 5), high fat-fed rats averaged 2.6 lesions per adenoma-bearing rat vs. 1.8 in the low fat-fed rats (*P* = 0.013 by extended Mantel-Haenszel test), a difference reflecting primarily experience with controls, who developed most of the adenomas. Control and OVX+E

TABLE 4

The effects of dietary fat and hormone treatment on the prevalence and total number of 7,12-dimethylbenz(a)anthracene-induced mammary adenocarcinomas in female Sprague-Dawley rats at necropsy¹

Diet	Control	OVX	OVX+E	Pooled
% of rats with an adenocarcinoma ²				
Beef tallow				
Low	59	9	47	39
High	63	16	84	54
Corn oil				
Low	41	6	47	31
High	69	16	75	53
Pooled	58	12	63	44
total number of adenocarcinomas ²				
Beef tallow				
Low	44	3	51	98
High	57	7	84	148
Corn oil				
Low	27	2	54	83
High	48	8	86	142
Pooled	176	20	275	471

¹Dietary fat, as either beef tallow or corn oil, was provided at 10 or 40% of energy. Hormone treatments were sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E). *n* = 32 rats per group.

²The odds of developing at least one adenocarcinoma increased a factor of 2.8 ± 0.8 (estimated odds ratio ± SE) in rats fed the high fat diets (*P* < 0.0001). Among rats showing at least one adenocarcinoma at necropsy, we observed no dietary effects on adenocarcinoma multiplicity. Control and OVX+E rats showed similar prevalences of adenocarcinomas at necropsy, although the number of adenocarcinomas per adenocarcinoma-bearing rat increased from 1.33 in the controls to 2.38 in the OVX+E rats (*P* = 0.0003 by Wald chi-square from a main effects logistic model).

rats showed similar prevalences of adenocarcinomas at necropsy, although the number of adenocarcinomas per adenocarcinoma-bearing rat increased from 1.33 in the controls to 2.38 in the OVX+E group (*P* = 0.0003 by Wald chi-square from a main effects logistic model). On the other hand, the odds of adenoma at necropsy in control rats were increased by a factor of 4.7 ± 1.4 relative to the OVX+E rats, but no effects of hormone treatment were observed on multiplicity.

Basal hormone concentrations. Basal levels of serum prolactin and estrogen from rats not treated with DMBA are shown in Figure 5, and pituitary weight, protein content and prolactin are shown in Figure 6 for rats at 24 wk of age, 16 wk after initiation of dietary treatment. We found no effects of dietary fat level or fat source on serum prolactin, serum estrogen or pituitary prolactin. Hormone status had a significant effect on serum prolactin and estrogen. Ovariectomy reduced serum prolactin and serum estrogen, compared with proestrus control or OVX+E rats. Diestrus control rats had lower serum prolactin

TABLE 5

The effects of dietary fat and hormone treatment on the incidence and total number of 7,12-dimethylbenz(a)anthracene-induced mammary adenomas in female Sprague-Dawley rats¹

Diet	Control	OVX	OVX+E	Pooled
% of rats with an adenoma ²				
Beef tallow				
Low	38	6	9	18
High	66	9	31	35
Corn oil				
Low	44	9	16	23
High	56	9	19	28
Pooled	51	9	19	26
total number of adenomas ²				
Beef tallow				
Low	22	2	8	32
High	51	3	19	73
Corn oil				
Low	26	4	8	38
High	68	4	12	84
Pooled	167	13	47	227

¹Dietary fat, as either beef tallow or corn oil, was provided at 10 or 40% of energy. Hormone treatments were sham ovariectomy (control), ovariectomy (OVX) and ovariectomy with estrogen implant (OVX+E). *n* = 32 rats per group.

²The odds of an adenoma at necropsy in control rats were increased by a factor of 4.7 ± 1.4 relative to OVX+E rats ($P < 0.0001$), but no effects of hormone treatment were observed on multiplicity. The odds of developing at least one adenoma were increased by a factor of 2.3 ± 0.7 (estimated odds ratio $\pm$ SE), in rats fed the high fat diets ($P = 0.01$). Those fed high fat diets averaged 2.6 lesions per adenoma-bearing rat vs. 1.8 in the low fat-fed rats ($P = 0.013$ by extended Mantel-Haenszel test).

and estrogen than proestrus control rats. Serum estrogen in OVX+E rats was more than double the proestrus levels of control rats. Ovariectomy with estrogen supplementation raised pituitary weight, protein content and prolactin content above their levels in the other hormone status groups ($P < 0.0001$), which were comparable to one another.

Prolactin metabolism. The immunoreactivity and radioactivity of the prolactin used for intravenous administration are shown in Figure 7. Following intravenous injection, collection of serum and subsequent chromatography, we observed three peaks of radioactivity (Fig. 8). The first peak eluted soon after the void volume and corresponded to the presumed dimeric form of circulating prolactin or monomeric prolactin bound to circulating proteins. The second component eluted at the position corresponding to the administered monomeric form of prolactin. The third component was the free iodine from the deiodination of the injected prolactin. There was no measurable immunoactivity by RIA in the fractions composing the third peak. The monomeric and larger

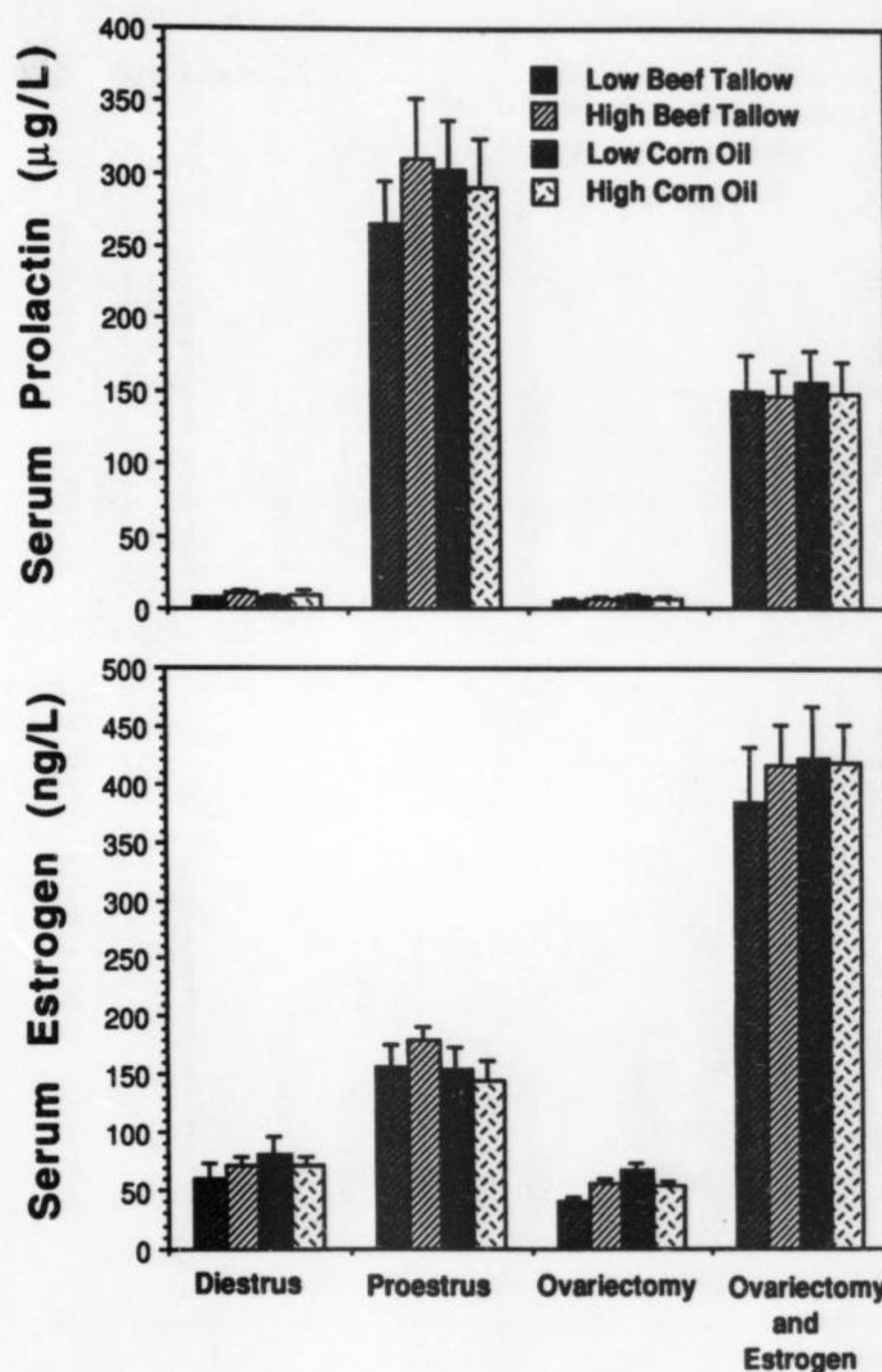


FIGURE 5 The effects of hormone status, dietary fat concentration and dietary fat source on serum prolactin and estrogen concentrations in female Sprague-Dawley rats at 24 wk of age, 16 wk after initiation of dietary treatment. Values represent means $\pm$ SEM. No effects of dietary fat level or fat source on serum prolactin and estrogen concentrations were observed. With the exception of ovariectomized rats vs. diestrus controls, all comparisons between groups paired according to ovarian status were significantly different for both serum prolactin and estrogen concentrations ($P < 0.0001$ from Scheffé's multiple comparison procedure).

forms of prolactin were significantly less at 20 min after injection compared with 10 min after injection, regardless of the method of measurement ($P < 0.0001$). Radioactivity in the combined monomeric and larger prolactin fractions decreased by 44% from 10 to 20 min. Total immunoreactivity decreased by 29% over the same period. No statistically significant effects of diet or hormone status were observed on total prolactin or the proportions in monomeric and larger forms at either time point. Examination of the radioactivity data shows that ~85% of the prolactin was in the monomeric form at 10 min and ~81% at 20 min. In contrast, monomeric prolactin accounted for

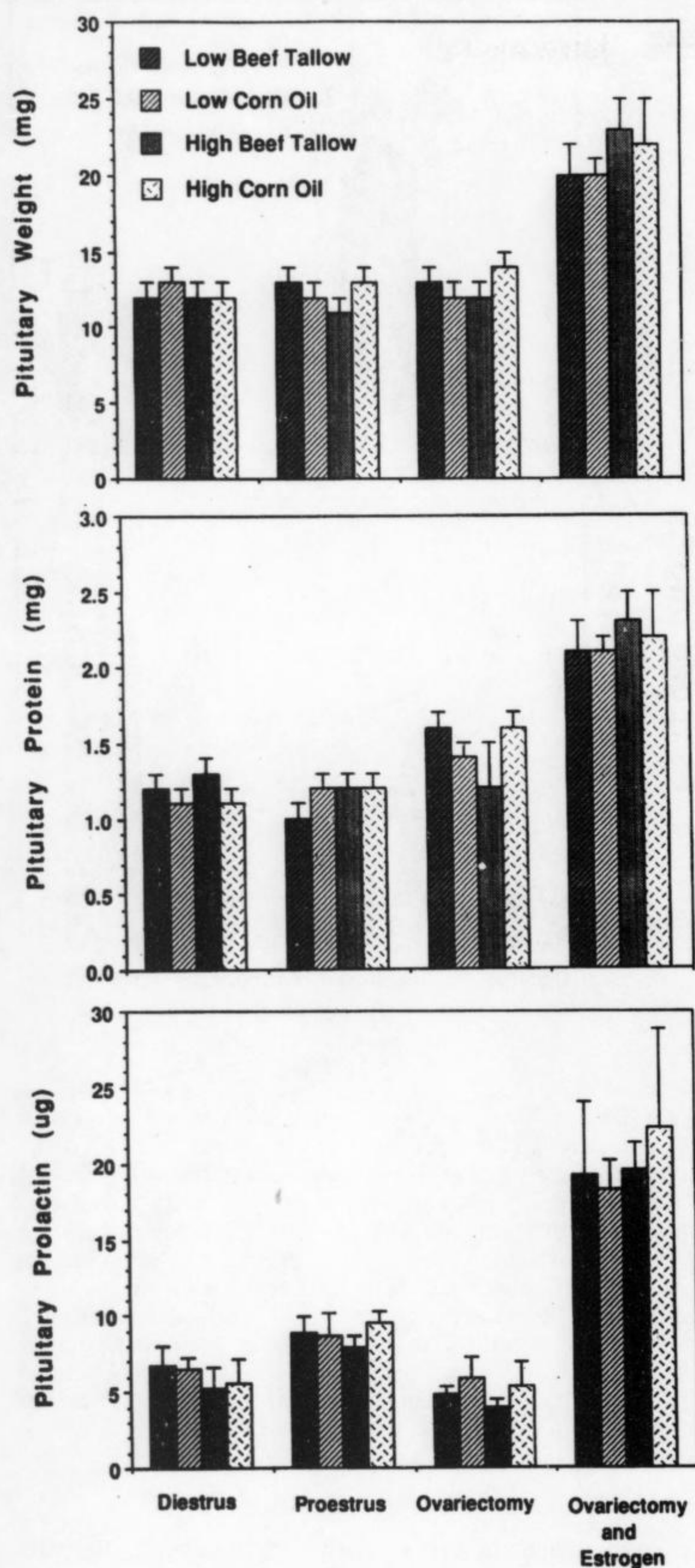


FIGURE 6 The effects of hormone status, dietary fat concentration and dietary fat source on pituitary weight, protein and prolactin in female Sprague-Dawley rats at 24 wk of age, 16 wk after initiation of dietary treatment. Values represent means $\pm$ SEM. Ovariectomized rats with estrogen implants showed greater pituitary weight, protein content and prolactin concentrations compared with other groups ($P < 0.0001$ from Scheffé's multiple comparison procedure).

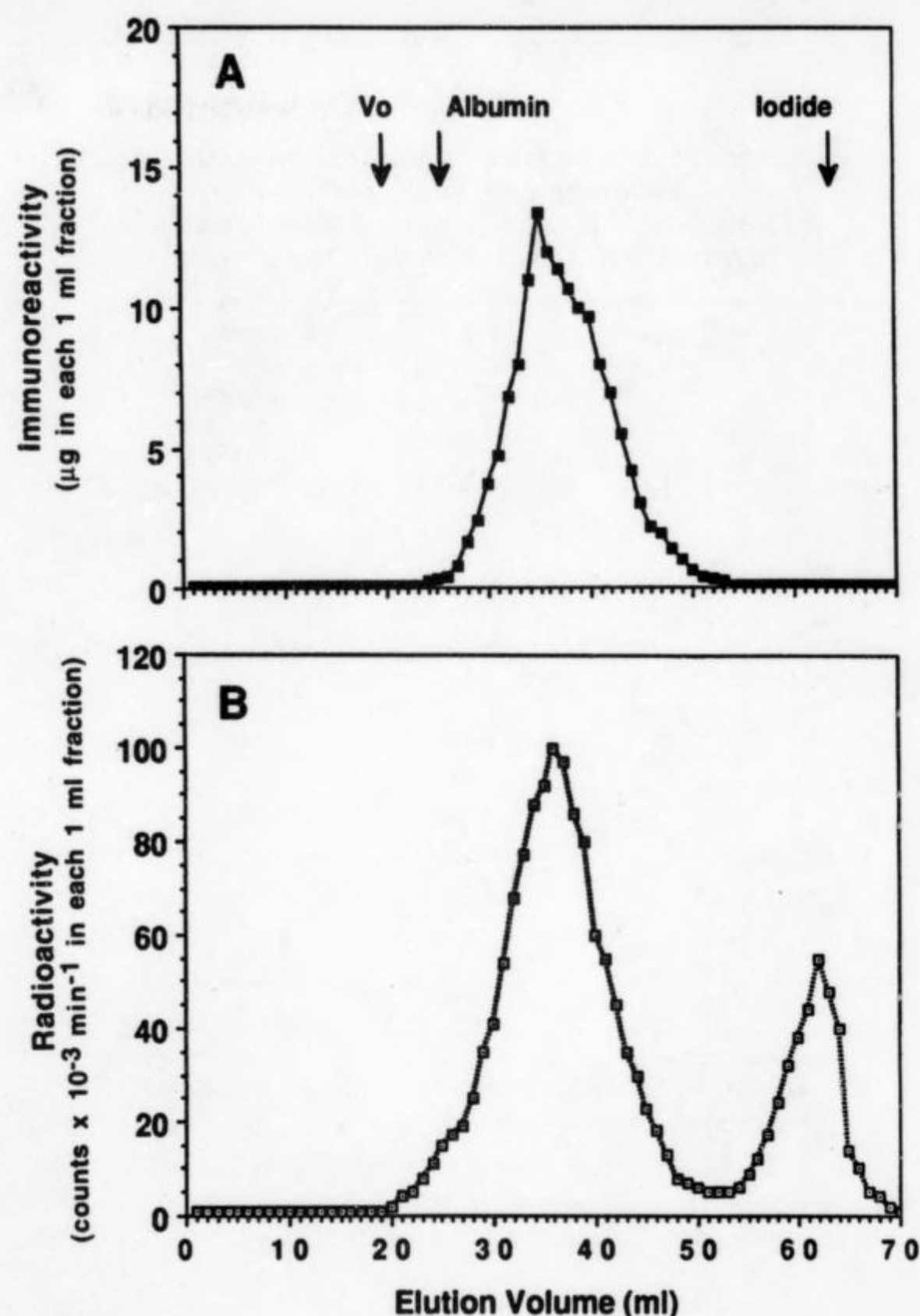


FIGURE 7 Chromatographic profile on Sephadex G-100 of ^{125}I -labeled rat prolactin used for intravenous injection into rats. The column was equilibrated and eluted at room temperature with phosphosaline buffer, pH 7.6, containing 2 g/L bovine serum albumin. Arrows indicate the positions of the void volume (V_0), peaks of human serum albumin and free ^{125}I . The immunoreactivity is expressed as the amount of prolactin detected by RIA in each 1-mL fraction eluted from the column (panel A). The radioactivity is expressed as the counts in each 1-mL fraction eluted from the column (panel B). The counts can be converted to becquerels (disintegrations per second) by multiplying values by 0.0183.

~58% of the radioimmunoassayable prolactin at 10 min, and ~63% at 20 min.

DISCUSSION

This study provides evidence that the cyclic function of the ovary is not necessary for dietary fat to increase the incidence of breast cancer. It is clear that fat can exert its stimulative effect on mammary carcinogenesis independently of the pituitary-ovarian axis. This conclusion is also supported by our data and that derived by other laboratories showing no significant differences in circulating estrogen concentrations during the estrous cycle in rats fed diets

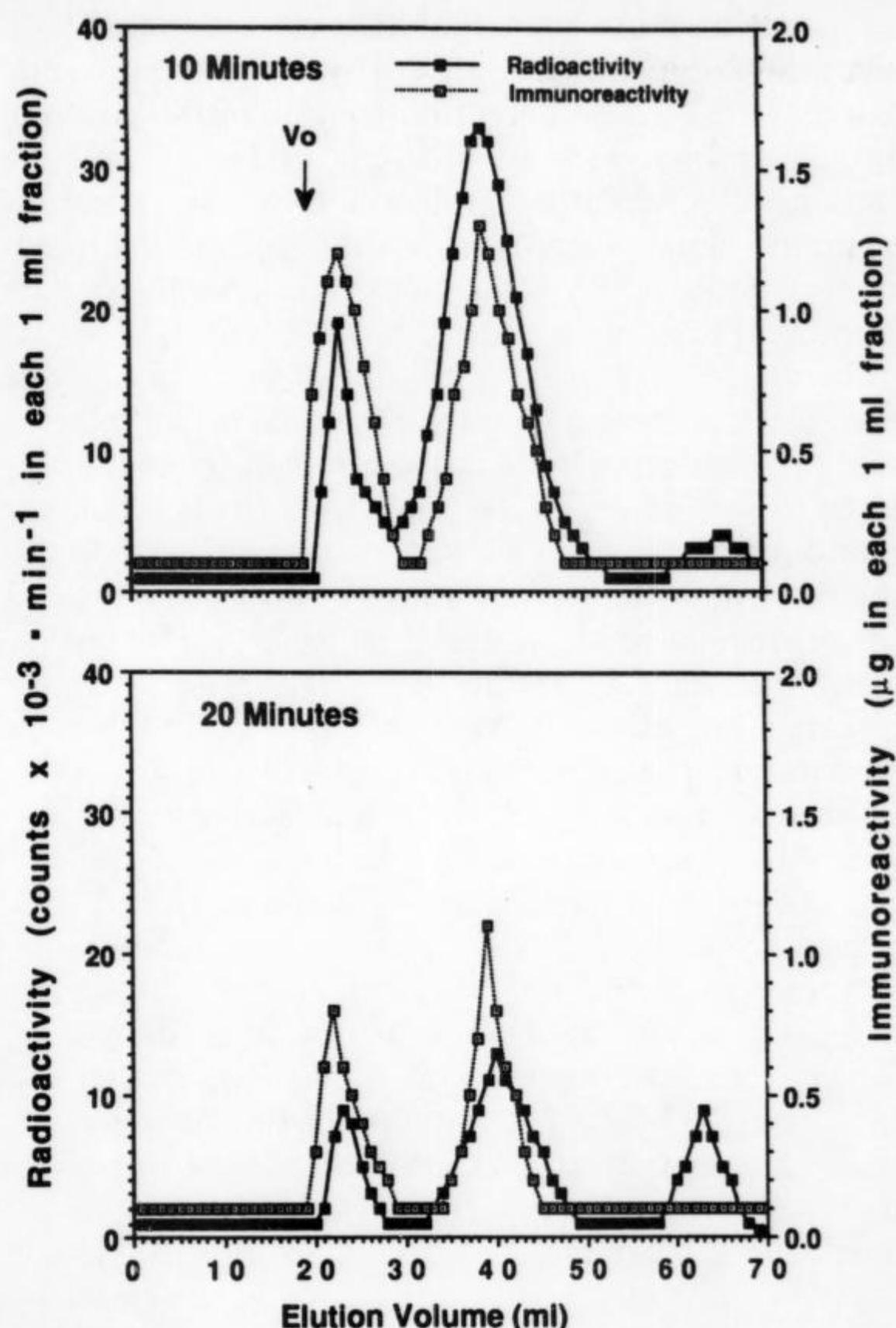


FIGURE 8 A representative prolactin profile following Sephadex G-100 of the sera from one rat at 10 and 20 min after injection with monomeric ^{125}I -labeled rat prolactin. Radioactivity counting and RIA were completed on each 1 mL fraction eluted from the column. The counts can be converted to becquerels (disintegrations per second) by multiplying values by 0.0183. The first peak that elutes soon after the void volume corresponds to the protein bound, dimeric or polymeric forms of circulating prolactin produced in vivo after administration of monomeric prolactin. The second component elutes at the position corresponding to the administered monomeric form of prolactin. The peak between fraction 60 and 70 (60 and 70 mL) corresponds to free iodide resulting from the deiodination for ^{125}I -labeled prolactin and shows no immunoreactivity. Profiles were obtained from six rats on each of 12 hormone by dietary treatment combinations. No significant effects of diet or hormone status were observed on total prolactin or the proportions in monomeric and larger forms at each time point. Averaged over all rats, radioactivity data shows that ~85% of the prolactin was in the monomeric form at 10 min and ~81% at 20 min. In contrast, monomeric prolactin accounted for ~58% of the radioimmunoassayable prolactin at 10 min and 63% at 20 min.

varying in fat over the range of intake commonly found in human populations (Ip and Ip 1981, Welsch 1987, Wetsel et al. 1984). Our findings are further supported by an earlier study (Aylsworth et al. 1984),

which was conceptually similar in design, but differed in scope and endpoints evaluated. In a 22-wk study, these authors had observed that ovariectomized and estrogen-treated rats developed larger tumors with a shorter latency when fed high fat diets, although no effects on tumor incidence were detected.

This study has important implications for future investigations designed to define the mechanism whereby fat enhances breast cancer risk. Our results and others suggest that fat enhances mammary tumorigenesis by modulating estrogen-regulated biological processes within the target tissue, rather than through an alteration in estrogen secretion by the ovary. Efforts should be directed toward a better understanding of how diet may modulate autocrine, paracrine and juxtacrine growth factor interactions and other cell signaling mechanisms among the mammary epithelia, stromal elements and vascular structures within the breast. Other studies have shown that dietary fat may alter the responsiveness of the mammary gland to estrogens or other steroids (Welsch 1985). Although very low levels of dietary fat (0.5 g/100 g of diet) may reduce estrogen receptor number, no differences have been reported comparing moderate to high levels of dietary fat (Ip and Ip 1981). However, enhanced estrogen-induced mammary gland growth has been observed in mice fed high fat (20 g/100 g of diet) compared with low fat (5 g/100 g of diet) diets (Welsch 1987). The precise estrogen responsive cellular signaling pathways and genetic programs that may be modulated by dietary fat in mammary tissue remain to be elucidated. To identify novel dietary fat-regulated genes in murine mammary tissue, our laboratory group is currently cloning genes from mammary tissue exhibiting higher or lower expression as lipid intake is increased. The recently cloned gene designated LFM1 (low fat mammary one) is down-regulated in mammary epithelial cells by diets high in fat (Elliot et al. 1993). We hypothesize that LFM1 may be one link between fat intake and the response of the mammary gland to tumor-promoting steroid hormones. Studies are currently underway to define the function of LFM1 and its role in the modulation of mammary carcinogenesis by dietary fat.

These studies do not completely eliminate the possibility that dietary fat alters the bioavailability of circulating estrogen, a hypothesis that has not been thoroughly investigated. However, a few studies in women have shown no effect of dietary fat on circulating sex hormone-binding globulin (Prentice et al. 1990, Rose et al. 1992, Shimizu et al. 1990). At the present time we can conclude that dietary fat stimulates experimental mammary carcinogenesis independently of ovarian steroid hormone secretion and has no significant effect on serum estrogen concentration. Our studies suggest that dietary components other than fat concentration may contribute to the lower

plasma estradiol concentrations seen in postmenopausal women after consumption of a low fat diet (Prentice et al. 1990).

The increase in mammary cancer incidence associated with the high fat diet was correlated with reduced survival. Most nutrition studies of chemically induced mammary cancer use relatively high doses of carcinogen, and are arbitrarily ended 20–40 wk after initiation. Such studies examine primarily the rate of tumor formation rather than the true incidence over the lifetime of the animals. The present study employed a relatively low dose of DMBA and was continued for 70 wk. Untreated female Sprague-Dawley rats have a median life span of 80–100 wk, and succumb primarily to spontaneous mammary cancer (Durbin et al. 1966). As a group, our DMBA-treated rats experienced earlier mortality. This was primarily due to progressive mammary carcinogenesis, although we also observed mortality that was probably related to DMBA-induced toxicity in multiple organ systems. Ovariectomy prolonged survival after DMBA administration, consistent with its effect in untreated rats (Durbin et al. 1966). On the other hand, survival of estrogen-treated rats in our study was greatly reduced, due primarily to mammary cancer but also to other causes such as pituitary adenomas. We also observed that increasing fat from 10 to 40% of energy was associated with decreased survival, an effect that was most prominent in the control rats. Dietary fat source significantly affected survival only in estrogen-deprived (OVX) rats.

The role of prolactin in experimental or human breast cancer has become more difficult to define with the realization that prolactin exhibits a high degree of structural polymorphism (Sinha 1992). Variants of prolactin may be synthesized as a result of differential mRNA splicing of the single known prolactin gene (Sinha 1992). In addition, prolactin may undergo various post-translational structural modifications, which further increase the heterogeneity of prolactin secreted in different pathophysiologic states (Sinha 1992). Circulating prolactin may bind to a variety of different proteins in serum, which in turn may modulate its bioactivity (Sinha 1992). Despite the complexities of prolactin homeostasis, our studies with dietary fat at 10 and 40% of energy confirm many previous reports showing no effect of diet on serum or pituitary immunoreactive prolactin concentrations (Clinton et al. 1985, Hopkins et al. 1981, Wetsel et al. 1984). In addition, our current studies show no effect of fat on serum immunoreactive prolactin in rats where the levels of estrogen have been controlled.

It is now established that serum prolactin concentrations defined by immunoassays may not directly correlate with prolactin biopotencies because the hormone circulates in forms having variable immunoreactivity and biological activity (Sinha 1992). Some studies suggest that the specific forms of

prolactin found in the circulation are associated with alteration of mammary cancer risk in mice (Sinha and Baxter 1979). The different circulating forms may represent the secretion of different prolactin variants, binding of circulating prolactin to serum proteins, systemic metabolism of prolactin or a combination of factors (Sinha 1992). Our study was designed to determine if dietary fat altered the metabolism of serum monomeric prolactin. We observed that ~15–20% of the administered monomeric radiolabeled prolactin was rapidly converted to larger forms in the rat as was observed in mice (Sinha and Baxter 1979). Direct radioactivity assays and RIA measurements of the relative proportions of the forms of prolactin revealed discrepancies between the two types of measurements in rats. Our results are very similar to those reported for mice (Sinha and Baxter 1979), where monomeric prolactin constituted 80% of the total serum prolactin determined by radioactivity, whereas with RIA it accounted for a maximum of ~51%. We also documented that dietary fat between 10 and 40% of energy had no influence upon the interconversion of prolactin forms or the clearance of circulating prolactin regardless of the method of detection. Overall, the vast majority of data suggests that fat does not modulate pituitary prolactin secretion or systemic metabolism (Clinton et al. 1985, Hopkins et al. 1981, Welsch 1985, Wetsel et al. 1984). However, conclusions must await the more complete characterization of prolactin variants secreted, their metabolism, immunoreactivity and biological potency in the mammary gland. As with estrogens, it can be hypothesized that diet may alter the sensitivity of the mammary gland to prolactin, thereby directly influencing tumorigenesis. Limited studies suggest that very low intakes of fat may reduce prolactin receptor number (Cave and Jurkowski 1984, Wetsel and Rogers 1984). Defining the effects of dietary fat on the different forms of the prolactin receptor (Kelly et al. 1992), the signal transduction pathways and gene expression in the mammary gland are important studies for the future.

The literature concerning dietary fat and rodent mammary cancer reveals variable effects of dietary fat on energy intake and body weight. The rats fed high fat diets in our 70-wk study consumed more energy and weighed more than those fed low fat diets. It is likely that energy intake contributed to the enhancement of mammary tumorigenesis observed in those fed high fat diets in this study (Clinton et al. 1984 and 1986). These diets were similar to those previously fed in our large breast cancer studies in which no effect of fat concentration was noted on energy intake or weight gain (Clinton et al. 1984). It is clear that subtle differences in the types of lipids utilized and their concentration may influence palatability and food intake. These issues must be considered in the interpretation of results from dietary fat and breast cancer studies in rodents.

We observed major effects of ovarian status on body weight and to a lesser degree, energy intake. Ovariectomy increased body weight, largely due to increased carcass fat based upon observation at necropsy. The increased weight occurred even though energy intake was the same as controls. Estrogen-supplemented rats showed significantly lower body weight, were very lean, and consumed only slightly less energy than controls. Although hormone status may directly influence metabolism, ovariectomy seemed to lower spontaneous physical activity whereas estrogen supplementation was associated with greater physical activity and energy expenditure.

The relevance of rodent studies to human breast cancer is often questioned. In our opinion, the rodent studies and human epidemiologic studies are generally in agreement. In many well-designed laboratory studies employing a diverse array of breast cancer models, an increase in the concentration of dietary fat has enhanced tumorigenesis. An examination of the larger rodent experiments (Clinton et al. 1984, 1986 and 1988) or a meta-analysis of rodent studies (Freedman et al. 1990) would suggest that a doubling of fat intake within the range observed in human populations will approximately double tumor risk. This relationship is similar to the correlation observed in geographic epidemiologic studies evaluating age-adjusted death rates from breast cancer and estimated fat intake (Carroll and Kohr 1975). Although a series of case-control studies in women found no clear association between fat intake and breast cancer, a recent overview analysis that included 4427 breast cancer patients and 6095 control subjects found a significant association in postmenopausal women (Howe et al. 1990). It is important to recognize that the majority of case-control studies were far too small to have sufficient power to detect the dietary fat and breast cancer relationship observed in rodent experiments or by international correlational studies (Freedman et al. 1993). A similar overview analysis of the few prospective cohort trials has not been done. The newly planned Women's Health Initiative, which will include a prospective analysis of the effects of a low fat dietary pattern on many health related endpoints, may provide important data concerning this issue (Freedman et al. 1993). In the absence of randomized intervention trials, further efforts to establish objective and reproducible biomarkers of fat intake and improved assessment tools of fat intake in diverse human populations should be encouraged.

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